

MINI GAME PROJECT

MINI GAME PROJECT – TIC TAC TOE

PROJECT OVERVIEW :

The Tic Tac Toe game is a simple console-based mini project developed using C++. The project is designed to demonstrate basic programming concepts such as loops, arrays, functions, and conditional statements through an interactive two-player game. Players take turns marking positions on a 3×3 game board using symbols X and O. The program checks for winning conditions after every move and declares the winner or draw result accordingly.

This project helps beginners understand game logic implementation, user interaction, and board management in C++ programming.

KEY FEATURES :

Console-based interactive Tic Tac Toe game.

Two-player gameplay support.

Dynamic display of the game board after every move.

Uses arrays to manage the game board.

Implements loops and conditional statements for game control.

Detects winning conditions for rows, columns, and diagonals.

Displays draw result when all positions are filled.

Simple and user-friendly interface.

TECHNOLOGIES USED :

Programming Language: C++

Concepts Used:

Arrays

Loops

Functions

Conditional Statements

User Input Handling

EXPECTED OUTCOME

The project provides an interactive gaming experience where two players can play Tic Tac Toe in the console. It successfully demonstrates the implementation of game logic, board updating, player switching, and win/loss detection using core C++ programming concepts.

CODE :

```
#include <iostream>
```

```
using namespace std;
```

```
char board[3][3] = {
```

```
    {'1','2','3'},
```

```
    {'4','5','6'},
```

```
    {'7','8','9'}
```

```
};
```

```
char player = 'X';
```

```
void showBoard() {
```

```
    cout << "\n";
```

```
    for(int i=0;i<3;i++) {
```

```
        for(int j=0;j<3;j++) {
```

```
            cout << " " << board[i][j];
```

```
            if(j<2) cout << " |";
```

```
        }
```

```
        cout << "\n";
```

```
        if(i<2) cout << "---|---|---\n";
```

```
    }
```

```
}
```

```
bool win() {
```

```
    for(int i=0;i<3;i++) {
```

```
        if(board[i][0]==player && board[i][1]==player && board[i][2]==player)
```

```
return true;
```

```
if(board[0][i]==player && board[1][i]==player && board[2][i]==player)
```

```
return true;
```

```
}
```

```
if(board[0][0]==player && board[1][1]==player && board[2][2]==player)
```

```
return true;
```

```
if(board[0][2]==player && board[1][1]==player && board[2][0]==player)
```

```
return true;
```

```
return false;
```

```
}
```

```
int main() {
```

```
int pos, moves = 0;
```

```
while(true) {
```

```
showBoard();
```

```
if(player=='X')
```

```
cout << "\nLasya (X), enter position: ";
```

```
else
```

```
cout << "\nPavani (O), enter position: ";
```

```
cin >> pos;
```

```
int r = (pos-1)/3;
```

```
int c = (pos-1)%3;
```

```
if(board[r][c]=='X' || board[r][c]=='O') {
```

```
    cout << "Invalid move!\n";
```

```
    continue;
```

```
}
```

```
board[r][c] = player;
```

```
moves++;
```

```
if(win()) {
```

```
    showBoard();
```

```
    if(player=='X')
```

```
        cout << "\nLasya Wins!";
```

```
    else
```

```
        cout << "\nPavani Wins!";
```

```
    break;
```

```
}
```

```
if(moves==9) {
```

```
    showBoard();
```

```
    cout << "\nGame Draw!";
```

```
    break;
```

```
}
```

```
player = (player=='X') ? 'O' : 'X';
```

```
}
```

```
return 0;
```

```
}
```

OUTPUT:

```
1 | 2 | 3
```

```
---|---|---
```

```
4 | 5 | 6
```

```
---|---|---
```

```
7 | 8 | 9
```

Lasya (X), enter position: 1

```
X | 2 | 3
```

```
---|---|---
```

```
4 | 5 | 6
```

```
---|---|---
```

```
7 | 8 | 9
```

Pavani (O), enter position: 5

```
X | 2 | 3
```

```
---|---|---
```

```
4 | O | 6
```

---|---|---

7 | 8 | 9

Lasya (X), enter position: 2

X | X | 3

---|---|---

4 | O | 6

---|---|---

7 | 8 | 9

Pavani (O), enter position: 8

X | X | 3

---|---|---

4 | O | 6

---|---|---

7 | O | 9

Lasya (X), enter position: 3

X | X | X

---|---|---

4 | O | 6

---|---|---

7 | O | 9

Lasya Wins!